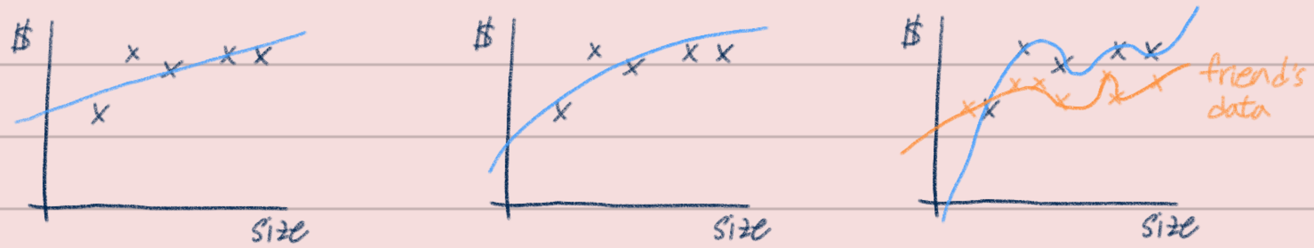
Lecture 8. Data Splits, Models & Cross-Validation

Outline

- Bias / Variance
- Regularization   one of the most effective way to prevent overfitting
- Train / dev / Test splits
- Model selection & cross validation

Bias / Variance



linear $\theta_0+\theta_1 x$ — underfit, high bias

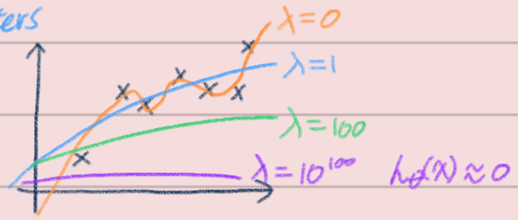
$\theta_0+\theta_1 x+\theta_2 x^2$ — "Just right"

$\theta_0+\theta_1 x+\theta_2 x^2+\cdots\ \theta_5 x^5$ — overfit, high variance

Regularization   by adding that penalty on the norm of parameters

let's take linear regression

$$\min_\theta \frac{1}{2}\sum_{i=1}^{m}\|y^{(i)}-\theta^T x^{(i)}\|^2+\underbrace{\boxed{\frac{\lambda}{2}\|\theta\|^2}}_{\text{regularization}}$$



Logistic regression

$$\arg\max_\theta \sum_{i=1}^{n}\log P(y^{(i)}|x^{(i)};\theta)\underbrace{\boxed{-\lambda\|\theta\|^2}}_{\text{regularization}} \longleftrightarrow \min\|w\|^2$$

SVM doesn't overfit data like crazy

∵ the objective of SVM is to minimize the norm of w squared (to maximize GM)

e.g. text classifier

데이터 샘플 수 $m=100$
단어 feature 수 $n=10000$

} regularization + logistic regression outperforms Naive Bayes
but without regularization, Logistic Regression overfits.
이때는 n should be $\le 10m$

Training set $S=\{(x^{(i)},y^{(i)})\}_{i=1}^{m}$

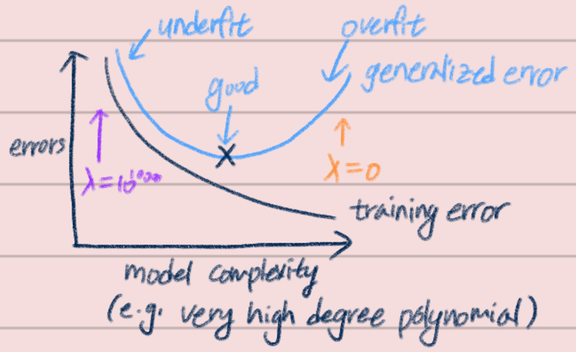
The most likely $\theta \rightarrow P(\theta|S)=\dfrac{P(S|\theta)\cdot P(\theta)}{P(S)} \rightarrow \arg\max_\theta P(\theta|S)=\arg\max_\theta P(S|\theta)\cdot P(\theta)$

$$\rightarrow \arg\max_\theta \Big(\underbrace{\prod_{i=1}^{m}P(y^{(i)}|x^{(i)};\theta)}_{\text{logistic regression}}\Big)P(\theta)$$

$P(\theta)$: $\theta\sim N(0,\tau^2 I)$ 가정시안

$$P(\theta)=\frac{1}{\sqrt{2\pi}(\tau^2 I)^{\frac{1}{2}}}\exp\left(-\theta^T(\tau^2 I)^{-1}\theta\right)$$

- Frequentist : $\underset{\theta}{\text{argmax}}\ P(S|\theta)$ — MLE
- Bayesians : prior distribution $P(\theta)$ $\quad \underset{\theta}{\max}\ P(\theta|S)$ — MAP (Maximum a posteriori)
  - ⇒ look at the data, compute the Bayesian posterior distribution of Theta, pick the value of θ that's most likely.



A few different mechanic procedures for trying to find the point in the middle:

e.g. 10,000 samples : polynomial $\overset{i=1}{\theta_0 + \theta_1 x}$? $\overset{i=2}{\theta_0 + \theta_1 x + \theta_2 x^2}$?

or $\overset{i=3}{\text{choose } \lambda}$ or $\overset{i=4}{\text{choose } \tau}$ or $\overset{i=5}{\text{choose } C}$

Training / Dev / Test Sets $\quad \overbrace{S_{train} \quad S_{dev}}^{70\%} \quad \overset{30\%}{S_{test}}$

$\qquad\qquad\qquad\qquad\qquad\quad$ 60% $\quad$ 20% $\quad$ 20% → data 아주 클 경우 90% 5% 5%

1. Train each model i (e.g. option for degree of polynomial) on S_{train} and Get some hypothesis function h.
2. Measure error on S_{dev}, Pick model with lowest error (squared error) on S_{dev} ← 여기서 S_{train} 쓰면 overfitting 이슈
3. Optional if evaluate algorithm on S_{test}, report the error from here
   ∵ 2번에서의 S_{dev} error는 valid unbiased model로 간주 X

(Simple) Hold-out cross validation (CV)

"development set" == "cross validation set"

Small datasets

$m = 100 \rightarrow 70\ S_{train},\ 30\ S_{dev}$?

k-fold CV $\quad k=5$ for illustration ($k=10$ is typical)

$$S = \begin{bmatrix} x^1, y^1 \\ \vdots \\ x^{100}, y^{100} \end{bmatrix}$$

For $i = 1 \ldots k$,
  Train (fit parameters) on $k-1$ pieces,
  Test on remaining 1 piece
} k개의 real numbers (squared errors)

Do Average

Avg( ) ⇒ shows how good a classifier with that degree polynomial is.

Final optional step : refit model on 100% of data leave-one-out CV $k = m \leq 50$ (very small data set)

Feature selection

One way to reduce overfitting is to find a small subset of the features that are most useful

가능한 feature 갯수는 a few hundred 지만 사실 critical 한건 몇개 안될때

Start with $F = \emptyset$ (empty set of features) $h(x) = \theta_0$

Repeat:

1) try adding each feature i to F, and see which single-feature addition most improves the dev set performance

2) add that feature to F

e.g. $[\emptyset$

$$\left[\begin{array}{ll} \emptyset + x_1 & h_\theta(x) = \theta_0 + \theta_1 x_1 \\ \emptyset + x_3 & \\ \emptyset + x_5 & h_\theta(x) = \theta_0 + \theta_1 x_5 \end{array}\right. \longrightarrow F = \{x_2\}$$

$$\left[\begin{array}{l} x_2 + x_1 \\ \vdots \\ x_2 + x_5 \end{array}\right. \longrightarrow F = \{x_2, x_4\}$$